

Comparative spectroscopic study of Dy³⁺ doped crystals and glasses for visible-infrared laser applications

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ABSTRACT

We report a comparative investigation of the optical properties of Dy³⁺ doped low-phonon crystals and glasses for potential applications in visible and mid-infrared solid-state laser development. All samples were excited at 450 nm for visible emissions in the 480 - 800 nm spectral region and at 910 nm for mid-infrared emissions spanning 2100 - 5000 nm. Direct optical excitation in the blue resulted in dominant yellow emission bands centered near 575 - 585 nm originating from the $^4F_{9/2} \rightarrow ^6H_{13/2}$ transition. In addition, weaker emission features were observed in the blue (~480 nm) and red (~670 - 750 nm) spectral regions corresponding to the $^4F_{9/2} \rightarrow ^6H_{15/2}$, $^4F_{9/2} \rightarrow ^6H_{11/2}$, and $^4F_{9/2} \rightarrow ^6H_{9/2}$ transitions. Among the investigated hosts, Dy³⁺:KPb₂Cl₅ exhibited the strongest and most spectrally distinct yellow emission, while Dy³⁺:InF₃ also demonstrated well-defined visible emission features. In contrast, Dy³⁺:BaF₂ and Dy³⁺:Ga₂Ge₅S₁₃ displayed broadened emission bands accompanied by increased background contributions. The peak emission cross sections for the yellow $^4F_{9/2} \rightarrow ^6H_{13/2}$ transition were determined to be 0.8×10^{-20} cm² for Dy³⁺:KPb₂Cl₅ and 0.1×10^{-20} cm² for Dy³⁺:InF₃. Fluorescence lifetime measurements of the $^4F_{9/2}$ excited state revealed millisecond-scale decay in Dy³⁺:KPb₂Cl₅ and Dy³⁺:InF₃, while significantly faster decay times (< 5 μs) were observed in Dy³⁺:BaF₂ and Dy³⁺:Ga₂Ge₅S₁₃. Mid-infrared emission from the $^6H_{11/2} \rightarrow ^6H_{13/2}$ transition was observed in the 3 - 5 μm spectral region under near-infrared excitation, exhibiting millisecond-scale lifetimes in both crystalline and glass hosts. The observed differences in spectral characteristics and decay dynamics are attributed to host-dependent phonon energy and local structural environment.

Keywords: rare earth ions, dysprosium, visible, crystals, glasses, solid-state lasers

1. INTRODUCTION

Solid-state laser sources operating in the visible to mid-infrared (mid-IR) spectral regions are of significant interest for a broad range of scientific, medical procedures, and defense-related applications [1-26]. In particular, yellow laser emission in the 575 - 585 nm range is highly desirable for applications including biomedicine, lidar, adaptive optics, and laser display technologies [3-12]. Similarly, mid-IR sources within the 3 - 5 μm atmospheric transmission window are critical for gas sensing, environmental monitoring, and remote sensing [13-26]. Dysprosium-doped materials have long attracted attention for visible emission and yellow solid-state laser development, with early demonstrations of Dy³⁺ laser action reported in tungstate crystals under flashlamp pumping and more recent room-temperature operation achieved using diode-pumped oxide and fluoride hosts [5-12]. Despite this progress, development of compact, efficient diode-pumped lasers in both the visible and mid-IR remains challenging due to strong non-radiative decay processes in conventional oxide and fluoride materials [7-12, 15-18]. Rare-earth doped low-maximum-phonon energy materials offer a promising pathway to mitigate multiphonon relaxation, enabling efficient visible and mid-IR emission at room temperature [4,12, 13-16]. In addition, the relatively broad absorption and emission features often observed in low-phonon halide and chalcogenide hosts can relax wavelength stabilization requirements for diode pumping and offer potential wavelength tunability.

Among rare-earth ions, dysprosium (Dy³⁺) is uniquely attractive due to its ability to support emission in both visible and mid-IR spectral regions using a single dopant [3-12, 14, 18-26]. The Dy³⁺ $^4F_{9/2} \rightarrow ^6H_{13/2}$ transition produces strong yellow emission under blue excitation, while the $^6H_{11/2} \rightarrow ^6H_{13/2}$ transition enables broadband mid-IR emission spanning the 3 - 5

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μm range. Previous studies have demonstrated efficient yellow emission and favorable spectroscopic properties in Dy^{3+} doped low-phonon chloride crystals, underscoring their potential for visible laser development [4]. However, practical implementation of these materials continues to involve considerations related to crystal growth and mechanical robustness. In this work, we present a comparative spectroscopic study of Dy^{3+} doped low-phonon crystals and glasses, using $\text{Dy}:\text{KPb}_2\text{Cl}_5$ as a reference system alongside BaF_2 single crystals as well as InF_3 and $\text{Ga}_2\text{Ge}_5\text{S}_{13}$ glass hosts. By systematically evaluating absorption characteristics, emission spectra, and decay dynamics for both visible yellow and mid-IR transitions, this study aims to identify host materials that balance strong spectroscopic performance with practical feasibility for solid-state laser applications across multiple spectral regions.

2. EXPERIMENTAL CONSIDERATIONS

Dy^{3+} doped low-phonon energy materials investigated in this study include KPb_2Cl_5 (KPC), BaF_2 single crystals and $\text{Ga}_2\text{Ge}_5\text{S}_{13}$ (GGS) and InF_3 glasses. The physical characteristics of these materials, including transparency ranges and maximum phonon energies, are summarized in Table 1. These hosts were selected to span a broad range of phonon energies and structural environments, enabling a comparative evaluation of Dy^{3+} spectroscopic behavior in both crystalline and amorphous low-phonon systems. Dy^{3+} doped single crystals were grown using the vertical Bridgman technique from high-purity starting materials and subsequently cut and polished to optical quality for spectroscopic measurements. Dopant concentrations, summarized in Table 2, were chosen to provide sufficient absorption strength while minimizing concentration quenching effects. Glass samples were prepared using conventional melt-quenching methods and annealed

Table 1. Physical characteristics of all studied Dy^{3+} doped single crystals and glasses [4,14,16-18].

Property	Crystals		Glasses	
	KPb_2Cl_5 (KPC)	BaF_2	$\text{Ga}_2\text{Ge}_5\text{S}_{13}$ (GGS)	InF_3
Crystal/Glass Structure	Monoclinic	Cubic	chalcogenide glass	fluoride glass
Transparency Range (μm)	0.3 - 15	0.2 - 14	0.48 - 10	0.2 - 9
Melting Point ($^{\circ}\text{C}$)	434	820	1150	1172
Density (g/cm^3)	4.6	4.2	4.58	5
Refractive index	2.01	~ 1.47	~ 2.15	~ 1.5
Maximal Phonon Energy (cm^{-1})	203	330	435	500

near the glass transition temperature to reduce internal stress and improve optical homogeneity. More details of the purification, crystal growth and fabrication processes were described elsewhere [4,14].

Room temperature transmission and absorption spectra were recorded using a Cary 5000 UV-Vis-NIR spectrophotometer and a Nicolet 6700 Fourier-transform infrared spectrometer. Visible emission spectra were obtained under excitation with a laser diode, while a continuous-wave Spectra-Physics Tsunami Ti:Sapphire laser ($\sim 905\text{ nm}$) was used for mid-IR emissions. A Horiba Fluorolog-3 system equipped with an iHR-320 monochromator and a 1200 grooves/mm grating blazed at 750 nm was used to acquire the visible emission spectra, whereas a Princeton Instruments Acton SpectraPro 0.15 m monochromator (λ_{blaze} : $4\text{ }\mu\text{m}$, 150 grooves/mm) was used to collect the mid-IR emission spectra. Emission signals were detected using either a thermoelectrically cooled photomultiplier tube (PMT) or a liquid-nitrogen-cooled InSb detector (Infrared Associates, $\sim 1.0 - 5.4\text{ }\mu\text{m}$) in conjunction with a Stanford Research Systems SR830 dual-phase lock-in amplifier. Fluorescence decay measurements were performed using the output of a pulsed (10 ns pulses, 10 Hz) Nd:YAG pumped Optical Parametric Oscillator (OPO) system. The emission decay transients were recorded with a LabView-driven National

Instruments data acquisition system. For temperature-dependent emission studies, samples were mounted on the cold finger of a two-stage closed-cycle helium refrigerator.

Table 2. Dy³⁺ dopant concentrations of all studied Dy³⁺ doped single crystals and glasses.

Materials	Dy ³⁺ Concentration (x 10 ²⁰ cm ⁻³)
Dy ³⁺ :KPb ₂ Cl ₅	0.50 [4]
Dy ³⁺ :BaF ₂	1.67 [18]
Dy ³⁺ :Ga ₂ Ge ₅ S ₁₃	2.51 [14]
Dy ³⁺ :InF ₃	0.20 [this work]

3. RESULTS AND DISCUSSION

Figure 1(a) shows the room-temperature transmission spectra of Dy³⁺ doped KPC and BaF₂ single crystals and Dy³⁺ doped GGS and InF₃ glasses. All materials exhibit high optical transmission across much of the visible and near-infrared spectral regions, with transmission levels typically exceeding ~60%. The absorption edge varies among the hosts, with the fluoride and chloride crystals extending further into the ultraviolet, while the glass samples exhibit earlier short-wavelength cutoffs associated with increased background absorption in amorphous hosts. The corresponding absorption spectra in Fig. 1(b) reveal well-defined Dy³⁺ ground-state absorption bands originating from the ⁶H_{15/2} manifold and terminating at higher-lying excited states, as illustrated in the energy level diagram in Fig. 2.

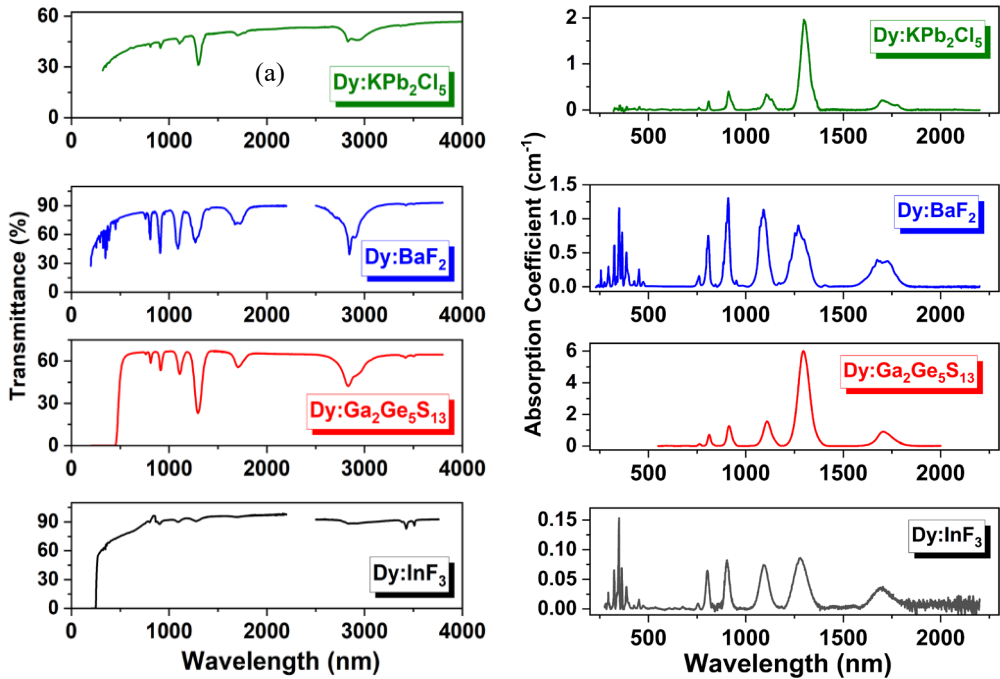


Figure 1. Room temperature (a) transmission (b) absorption spectra of Dy³⁺:KPb₂Cl₅ and Dy³⁺:BaF₂ single crystals, Dy³⁺:Ga₂Ge₅S₁₃, and Dy³⁺:InF₃ glasses.

Strong absorption features are observed in the visible region, with additional weaker bands extending into the near-infrared. The crystalline hosts display sharper and more structured absorption lines, whereas the glass hosts exhibit broader features due to the disordered local environments surrounding Dy^{3+} ions. These transmission and absorption characteristics confirm good optical quality of the investigated materials and identify suitable excitation bands for subsequent visible and mid-infrared emission studies [4,14,18].

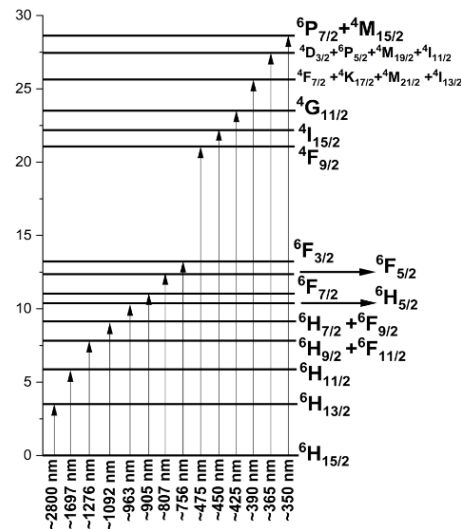


Figure 2. Energy level diagram showing the corresponding absorption lines from the ground state $6\text{H}_{15/2}$.

3.1 Yellow emission studies

Under blue excitation ($\lambda_{\text{ex}} \sim 450$ nm), all Dy^{3+} doped KPC, BaF_2 , and InF_3 exhibit emission originating from the $4\text{F}_{9/2}$ excited state, as shown in Fig. 3 (a). The dominant yellow emission band centered near 575 - 585 nm corresponds to the $4\text{F}_{9/2} \rightarrow 6\text{H}_{13/2}$ transition. Additional weaker emission bands are observed in the blue-green and red spectral regions and are assigned to the $4\text{F}_{9/2} \rightarrow 6\text{H}_{15/2}$ (478 nm), $4\text{F}_{9/2} \rightarrow 6\text{H}_{11/2}$ (665 nm), and $4\text{F}_{9/2} \rightarrow 6\text{H}_{9/2}$ (750 nm) transitions, respectively, consistent with the Dy^{3+} energy level scheme shown in Fig. 3 (b). Similar emission behavior for $\text{Dy}^{3+}:\text{KPC}$, including strong yellow emission accompanied by weaker blue and red bands, was previously reported by Hommerich et al. [4], with the favorable visible emission properties attributed to the low-phonon-energy chloride host lattice.

In $\text{Dy}^{3+}:\text{KPC}$ and $\text{Dy}^{3+}:\text{InF}_3$, the emission spectra are characterized by relatively sharp and well-resolved spectral features, indicating reduced inhomogeneous broadening and favorable local environments for radiative relaxation. In contrast, $\text{Dy}^{3+}:\text{BaF}_2$ and $\text{Dy}^{3+}:\text{GGs}$ exhibit broader emission features with a pronounced underlying background. For $\text{Dy}^{3+}:\text{GGs}$, this behavior is influenced in part by the short-wavelength absorption cutoff near ~ 480 nm, which limits excitation efficiency at 450 nm and introduces increased host-related background contributions. Although the phonon energies of BaF_2 (330 cm^{-1}) and GGs (435 cm^{-1}) are not significantly higher than those of $\text{Dy}^{3+}:\text{KPC}$ (203 cm^{-1}), the observed spectral broadening, and additional non-radiative relaxation pathways beyond multiphonon processes alone. Among the investigated hosts, $\text{Dy}^{3+}:\text{KPC}$ displays the strongest and most spectrally distinct yellow emission, consistent with the influence of its low-phonon-energy lattice on suppressing non-radiative decay.

Figure 4 (a) and (b) show room temperature fluorescence decay transients of the $4\text{F}_{9/2}$ excited state and the temperature dependence of the corresponding emission lifetimes for selected Dy^{3+} -doped hosts under blue excitation ($\lambda_{\text{ex}} \sim 450$ nm, $\lambda_{\text{mon}} \sim 570$ nm). $\text{Dy}^{3+}:\text{KPC}$ and $\text{Dy}^{3+}:\text{InF}_3$ exhibit long, nearly single-exponential decay with lifetimes of ~ 1.3 ms and 0.3 ms, respectively, indicating efficient suppression of non-radiative losses in these low-phonon hosts. In contrast, $\text{Dy}^{3+}:\text{BaF}_2$ and $\text{Dy}^{3+}:\text{GGs}$ display significantly faster decay dynamics with lifetimes shorter than $5 \mu\text{s}$. The emission lifetimes show no significant dependence on temperature between 10 K and room temperature for $\text{Dy}^{3+}:\text{KPC}$ and $\text{Dy}^{3+}:\text{InF}_3$, suggesting that multiphonon relaxation processes remain weak over this temperature range. This behavior is consistent with the large

energy gap ($\sim 7500 \text{ cm}^{-1}$) between the $^4F_{9/2}$ emitting level and the next lower-lying excited state, which suppresses thermally activated non-radiative decay.

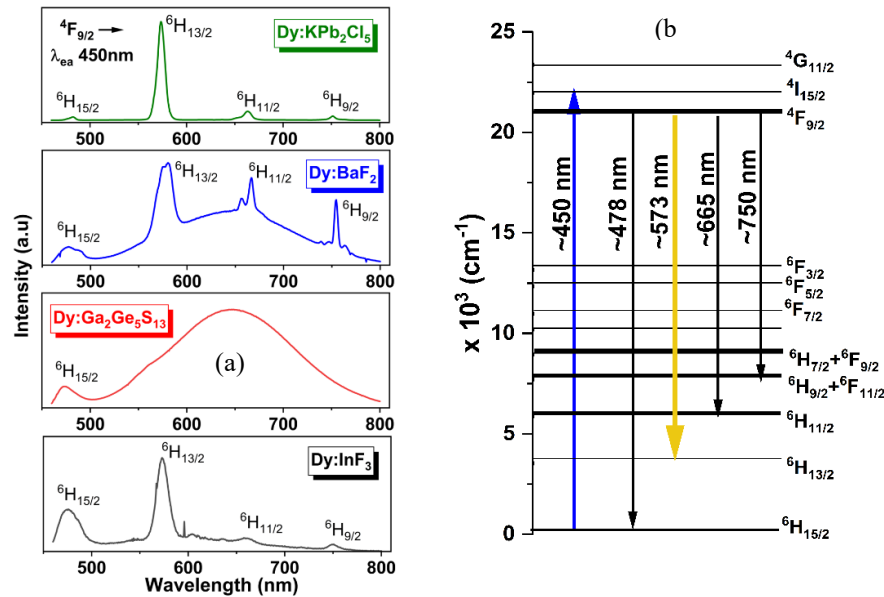


Figure 3. (a) Room-temperature visible emission spectra of Dy³⁺ doped studied materials recorded under 450 nm excitation. (b) Corresponding Dy³⁺ energy level diagram showing the excitation and emission transitions.

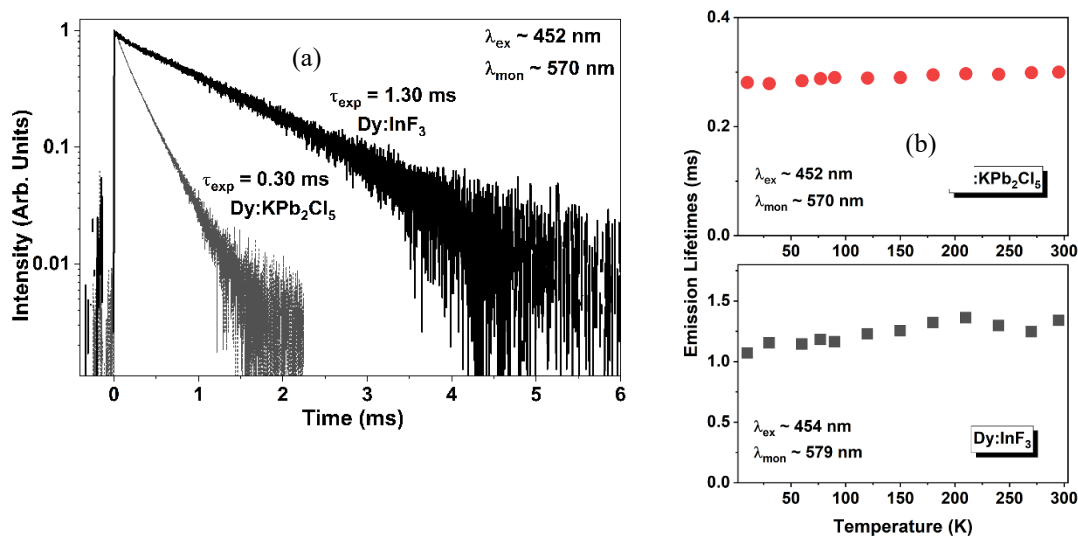


Figure 4. (a) Emission decay transients of the Dy³⁺ $^4F_{9/2} \rightarrow ^6H_{13/2}$ transition in Dy³⁺:KPb₂Cl₅ and Dy:InF₃ under blue excitation. (b) Temperature-dependent emission lifetimes measured for the same samples between 10 K and room temperature.

The unusually long lifetime observed in Dy³⁺:InF₃ may also reflect reduced radiative transition probabilities associated with a higher effective local site symmetry of Dy³⁺ ions in the fluorindate glass host, reflecting the distribution of local environments inherent to amorphous materials. Such symmetry-induced weakening of electric-dipole transition strengths is further supported by the comparatively small absorption coefficients and the low emission cross section obtained for

Dy³⁺:InF₃, as discussed below. For Dy³⁺:GGS, the extracted decay represents an effective lifetime under the present excitation and detection conditions, due to the short-wavelength absorption cutoff near ~480 nm and the increased contribution of host-related background discussed above. Accordingly, further discussion of decay mechanisms is focused on Dy³⁺:BaF₂, where Dy³⁺ emission features are clearly observed but fast decay dynamics persist. The markedly shorter lifetime observed in Dy³⁺:BaF₂ cannot be explained solely by host phonon energy, since BaF₂ possesses a relatively low maximum phonon energy (~330 cm⁻¹). Instead, the fast decay behavior is likely influenced by host-dependent effects such as crystal field interactions, defect- or impurity-related quenching, and possible energy transfer or cross-relaxation processes. Using the measured emission spectra and fluorescence lifetimes, the peak emission cross sections for the yellow ⁴F_{9/2} → ⁶H_{13/2} transition were calculated using the Fuchtbauer–Ladenburg relation and determined to be 0.8 × 10⁻²⁰ cm² for Dy³⁺:KPC [4] and 0.1 × 10⁻²⁰ cm² for Dy³⁺:InF₃. These results highlight the critical role of host lattice properties beyond phonon energy alone in governing Dy³⁺ excited-state dynamics and further support Dy³⁺:KPC crystal and Dy³⁺:InF₃ glass as promising host materials for visible emission and potential solid-state laser applications.

3.2 Mid-IR emission studies

Under excitation at ~905 nm, mid-infrared emission was investigated for all Dy³⁺-doped samples in the 2.1 - 5.0 μm spectral region. As shown in Fig. 5(a), all hosts exhibit an emission band centered near ~2.9 μm corresponding to the ⁶H_{13/2} → ⁶H_{15/2} transition. Figure 5(b) illustrates the Dy³⁺ energy level scheme associated with the observed mid-IR emission. A pronounced mid-IR emission band near ~4.4 - 4.5 μm, attributed to the ⁶H_{11/2} → ⁶H_{15/2} transition, is clearly observed in Dy³⁺:KPC and weakly in Dy³⁺:GGS, while no measurable mid-IR emission is detected from Dy³⁺:BaF₂ and Dy³⁺:InF₃ under identical excitation conditions. The absence of observable emission in BaF₂ and InF₃ highlights the importance of sufficiently low host phonon energy for sustaining mid-IR radiative transitions [17,18], given the relatively small energy gap (~2200 cm⁻¹) associated with the ⁶H_{11/2} upper laser level.

Figure 5(c) presents the fluorescence decay transients of the Dy³⁺ ⁶H_{11/2} excited state, yielding lifetimes of approximately 6.2 ms for Dy³⁺:KPC and 1.06 ms for Dy³⁺:GGS [14,25]. The corresponding peak emission cross sections for the ⁶H_{11/2} → ⁶H_{15/2} transition were calculated using the Fuchtbauer–Ladenburg relation based on the measured emission spectra and fluorescence lifetimes, yielding values of 1.27 × 10⁻²⁰ cm² for Dy³⁺:KPC and 1.05 × 10⁻²⁰ cm² for Dy³⁺:GGS. The mid-IR spectroscopic parameters for Dy³⁺:GGS were adopted from previously published work by the authors [14], while Dy³⁺:KPC was used as a reference sample based on prior reported results [25, 26] and re-evaluated under the same experimental conditions for direct comparison in the present study. These results underscore the importance of host

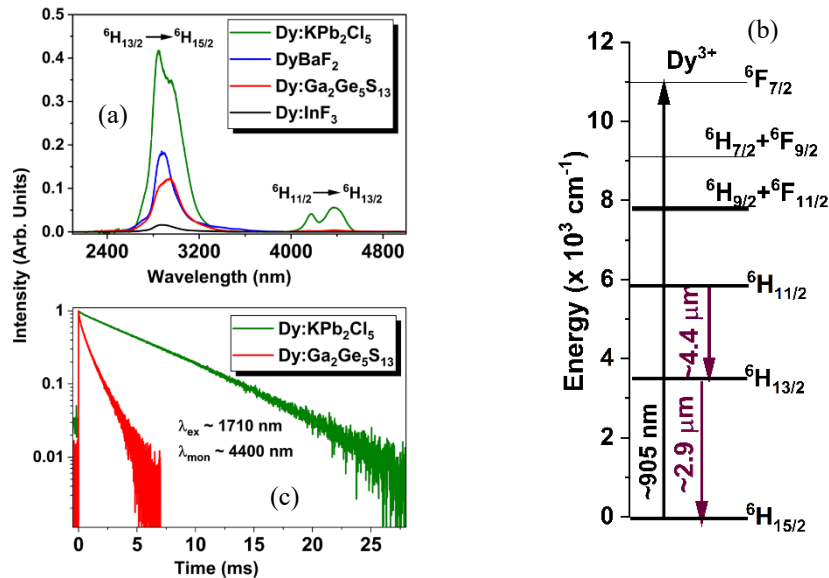


Figure 5. Mid-IR emission spectra and decay dynamics of Dy³⁺ doped low-phonon hosts under ~905 nm excitation: (a) emission spectra showing the ⁶H_{13/2} → ⁶H_{15/2} (~2.9 μm) and ⁶H_{11/2} → ⁶H_{15/2} (~4.4 - 4.5 μm) transitions, (b) corresponding Dy³⁺ energy level scheme, and (c) decay transients of the ⁶H_{11/2} level.

selection for enabling Dy^{3+} mid-IR emission in the 4 - 5 μm region: while Dy^{3+} :KPC exhibits the most favorable spectroscopic performance due to its very low phonon energy, practical considerations related to chloride crystal growth and mechanical robustness remain challenging. In contrast, Dy^{3+} :GGS glass, although exhibiting shorter lifetimes and weaker emission, offers greater fabrication flexibility and still demonstrates reasonable mid-IR spectroscopic properties, indicating that glass hosts remain viable alternatives for certain mid-infrared laser applications.

4. SUMMARY

A comparative spectroscopic evaluation of Dy^{3+} doped low-phonon crystals and glasses was performed to assess their potential for visible and mid-infrared solid-state laser applications. Under blue excitation, Dy^{3+} : KPb_2Cl_5 and Dy^{3+} : InF_3 exhibited the strongest and most spectrally distinct yellow emission from the $^4\text{F}_{9/2} \rightarrow ^6\text{H}_{13/2}$ transition, accompanied by millisecond-scale lifetimes and favorable emission cross sections, whereas Dy^{3+} : BaF_2 and Dy^{3+} : $\text{Ga}_2\text{Ge}_5\text{S}_{13}$ showed broadened emission features and fast decay dynamics. Mid-infrared emission under ~ 905 nm excitation revealed pronounced 4 - 5 μm emission only in Dy^{3+} : KPb_2Cl_5 , with a lifetime of 6.2 ms and a peak emission cross section of $1.27 \times 10^{-20} \text{ cm}^2$, while Dy^{3+} : $\text{Ga}_2\text{Ge}_5\text{S}_{13}$ exhibited weaker emission with shorter lifetime and reduced cross section; no measurable mid-IR emission was detected in Dy^{3+} : BaF_2 or Dy^{3+} : InF_3 . Although Dy^{3+} : KPb_2Cl_5 demonstrates the most favorable spectroscopic performance due to its very low phonon energy, challenges associated with chloride crystal growth and handling remain, whereas Dy^{3+} : $\text{Ga}_2\text{Ge}_5\text{S}_{13}$ glass offers greater fabrication flexibility with reasonable mid-IR emission characteristics. These results highlight complementary roles for crystalline and glass low-phonon hosts and motivate continued material optimization and laser demonstration studies for future visible and mid-infrared solid-state laser development.

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